Empirical Turnover Invariance in a Viability-Constrained Memory Substrate

VCSM Theory Series — Paper 8

2026

Abstract

We report an empirical turnover invariance in a viability-constrained adaptive memory substrate: the turnover rate that maximizes spatial structure formation, $\nu^* \approx 0.001$, is invariant to propagation strength κ across its full tested range (0 to 0.15, a > 150-fold variation), within this substrate and input protocol. Three supporting experiments confirm that ν^* is similarly stable across a 4× range of the consolidation gate (`SS`) and consolidation rate (`FIELD_ALPHA`) within their operational regimes. Taken together, the experiments reveal a clean separation of concerns: *adaptation dynamics are local* (the optimal turnover is determined by wave-grid interaction, not by the propagation or consolidation machinery), while propagation controls only the amplitude and spatial reach of learned structure. This separation suggests the adaptive regime location is an intrinsic property of the input protocol rather than the learning parameters, and motivates a broader hypothesis about local adaptation in systems with mandatory turnover.

1 Introduction

Paper 7 (Regime Structure) mapped the (ν, κ) parameter space of a VCML substrate and identified a bounded adaptive window: spatial structure is maximized at intermediate turnover ($\nu \in [0.001, 0.010]$) and is verified by three independent robustness experiments.

The key question this paper addresses is not *how robust* the window is, but *what sets its location*. Understanding the location law matters for design: if ν^* scales with κ, one would tune propagation to shift the learning regime. If it scales with `SS`, one would change the consolidation gate. If neither controls it, the regime boundary is set elsewhere—and that elsewhere is informative.

Main result. $\nu^* \approx 0.001$ is invariant to κ (Experiment 1, Figure 1). Three supporting sweeps of `SS`, `FIELD_ALPHA`, and `WAVE_RATIO` confirm the same location stability within practical operating ranges. The regime boundary is set by the copy-forward/erasure balance at the wave–grid level, and consolidation parameters control only the amplitude of structure formed within the window.

Architectural interpretation. The invariance of ν^* to κ implies that *learning is local*: the condition for effective adaptation—the turnover/wave balance—does not require the system to coordinate globally through propagation. Propagation then plays a complementary role: it spreads locally formed structure globally, amplifying what the local learning rule has already established. This maps onto a recurring design pattern in biological adaptive systems, discussed in Section 7.4.

1.1 Background: VCSM Consolidation Gate

In VCSM, a site updates its `fieldM` vector only after accumulating `SS` consecutive unperturbed steps:

$$\mathrm{ok}[i] \;=\; [\texttt{streak}[i] \geq \texttt{SS}] \tag{1}$$

When ok[i] holds:

$$\texttt{fieldM}[i] \;\; += \;\; \texttt{FIELD_ALPHA} \cdot (\texttt{mid_mem}[i] - \texttt{fieldM}[i]) \tag{2}$$

The consolidation timescale is thus jointly governed by SS (gate threshold, steps) and FIELD_ALPHA (learning rate per calm step). The probability that a site achieves a calm streak $\geq$ SS is approximately:

$$P_{\text{calm}}(\texttt{SS}) \;\approx\; \left(1 - \frac{\texttt{WR}}{2 \cdot \texttt{WAVE_DUR}}\right)^{\texttt{SS}+\texttt{WAVE_DUR}-1} \tag{3}$$

This baseline formula is relevant for understanding amplitude effects, but—as the experiments show—it does not control the location of ν^*.

2 Shared Protocol

All experiments use the Paper 7 Exp1 base configuration. Only the swept parameter varies; all others are fixed at reference values.

Parameter	Reference value
Grid	$W = 40$, $H = 20$, HALF $= 20$, $N_{\text{act}} = 400$
HS	2
KAPPA	0.020
WAVE_RATIO	4.8
WAVE_DUR	15
SS	10
FIELD_ALPHA	0.16
SEED_BETA	0.25
STEPS	2000
SHIFT	1000
Seeds	5
ν sweep	$\{0.0001, 0.0003, 0.001, 0.002, 0.005, 0.010, 0.020, 0.040, 0.080\}$

For each configuration we report ν^* (argmax sg4, 5-seed mean) and sg4* (the peak value). The key question in each experiment is whether ν^* shifts; amplitude changes are secondary.

3 Experiment 1: Propagation Invariance (Key Result)

3.1 Protocol

Re-analysis of Paper 7 Experiment 3 results (the $\nu \times \kappa$ grid, 5 seeds per cell). For each $\kappa \in \{0.000, 0.005, 0.020, 0.060, 0.150\}$ we identify $\nu_{\text{opt}} = \text{argmax}_\nu \; \texttt{sg4}(\nu, \kappa)$ using 5-seed means and the ν sweep common to Paper 7 Exp3: $\{0.0003, 0.001, 0.005, 0.015, 0.050\}$.

3.2 Results

Result. $\nu_{\text{opt}} = 0.001$ for *all* κ values tested—a > 150-fold range of propagation strength, from zero diffusion to strong coupling. The collapse boundary is also unchanged: ν_{crit} falls between 0.005 and 0.015 at every κ. Propagation κ shifts only the amplitude of sg4*: from 396.7 at $\kappa = 0$ to 360.9 at $\kappa = 0.15$, a modest $\sim$9% reduction.

Table 1: ν_{opt} and $\texttt{sg4}_{\text{opt}}$ by κ. The sg4 profile lists 5-seed mean sg4 at $\nu \in \{0.0003, 0.001, 0.005, 0.015, 0.050\}$, with the peak bolded. Collapse boundary (ν_{crit}) is between 0.005 and 0.015 for all κ.

κ	ν_{opt}	$\texttt{sg4}_{\text{opt}}$	sg4 profile					
0.000	0.0010	396.7	279	**397**	99	11	1
0.005	0.0010	383.1	271	**383**	88	9	1
0.020	0.0010	376.3	273	**376**	80	8	1
0.060	0.0010	375.1	273	**375**	68	6	1
0.150	0.0010	360.9	276	**361**	49	5	0

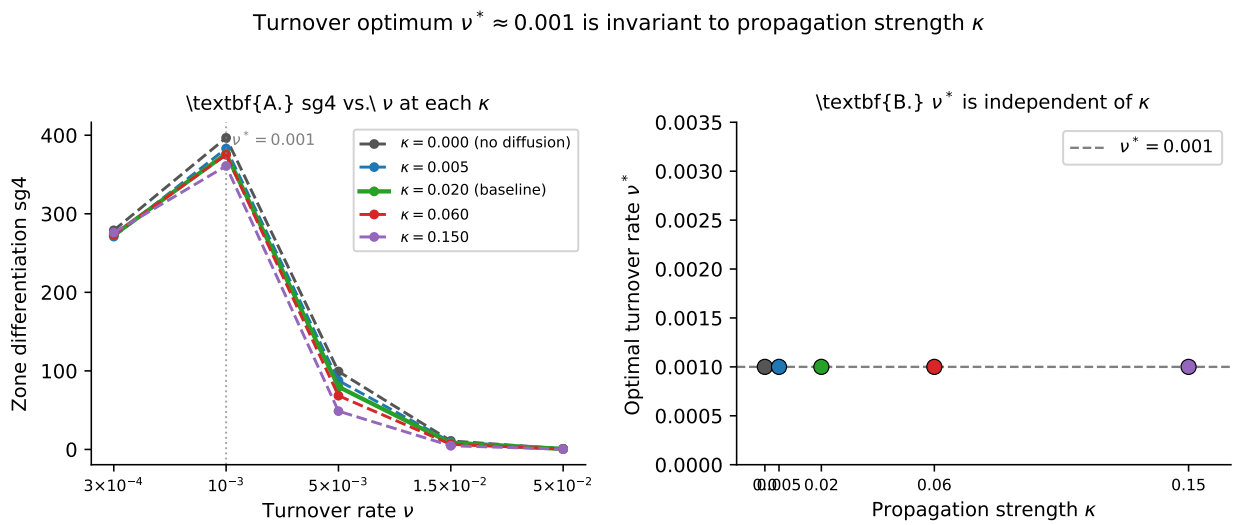

Figure 1: **The turnover law. A**: sg4 vs. ν at each κ value (Paper 7 Exp3 data). All five curves peak at the same $\nu^* = 0.001$ regardless of propagation strength. **B**: ν^* vs. κ—a horizontal line. Propagation strength does not set the adaptive regime location.

Figure 1 makes the invariance visually explicit. Panel A shows all five sg4-vs.-ν curves (one per κ) superimposed: the curves differ in height but peak identically at $\nu = 0.001$. Panel B plots ν^* vs. κ directly—a horizontal line.

4 Experiment 2: Consolidation Gate (Amplitude, Not Location)

4.1 Protocol

$\texttt{SS} \in \{5, 10, 20, 40\}$. All other parameters at reference. $4 \times 9 \times 5 = 180$ runs. The timescale-matching prediction is $\nu^* \propto 1/\texttt{SS}$ (inverse proportionality).

4.2 Results

Result. $\nu^* = 0.001$ for $\texttt{SS} \in \{5, 10, 20\}$, a 4× range. The timescale-matching prediction ($\nu^* \propto 1/\texttt{SS}$) fails: location is insensitive to SS. What *does* change is the amplitude: $\texttt{sg4}^*$ drops from 888 → 376 → 186 → 23 as SS doubles from 5 to 40, roughly consistent with the exponential decay of P_{calm} (Eq. (3)). At $\texttt{SS} = 40$, the gate is so strict that almost no site achieves 40 consecutive calm steps,

Table 2: SS sweep: ν^* and $\texttt{sg4}^*$ by consolidation gate. sg4 profile: $\nu = 0.0001, 0.0003, \ldots, 0.080$ (left to right).

SS	ν^*	$\texttt{sg4}^*$	sg4 profile								
5	0.0010	887.8	120	576	**888**	734	126	18	5	1	0
10	0.0010	376.3	223	273	**376**	328	80	24	4	1	0
20	0.0010	186.2	78	88	**186**	137	23	12	1	0	0
40	0.0020	23.5	0	0	0	**24**	5	0	0	0	0

collapsing structure entirely. The invariance therefore holds within the *operational consolidation regime* ($\texttt{SS} \leq 20$ at reference wave density); at extreme gate values, the system degrades into near-zero structure across all ν rather than shifting ν^* predictably.

5 Experiment 3: Consolidation Rate (Amplitude, Not Location)

5.1 Protocol

$\texttt{FIELD_ALPHA} \in \{0.04, 0.08, 0.16, 0.32\}$. All other parameters at reference. $4 \times 9 \times 5 = 180$ runs.

5.2 Results

Table 3: FIELD_ALPHA sweep: ν^* and $\texttt{sg4}^*$.

FIELD_ALPHA	ν^*	$\texttt{sg4}^*$	sg4 profile								
0.04	0.0001	188.0	**188**	124	174	136	32	11	2	0	0
0.08	0.0010	263.1	213	191	**263**	226	52	19	4	1	0
0.16	0.0010	376.3	223	273	**376**	328	80	24	4	1	0
0.32	0.0010	507.4	255	379	**507**	417	103	25	5	1	0

Result. $\nu^* = 0.001$ for $\texttt{FIELD_ALPHA} \in \{0.08, 0.16, 0.32\}$, a $4\times$ range. Amplitude scales monotonically: $\texttt{sg4}^*$ grows from $263 \to 376 \to 507$ roughly proportionally to FIELD_ALPHA in this range, confirming that the consolidation rate controls the height of the adaptive peak rather than its location.

At $\texttt{FIELD_ALPHA} = 0.04$, ν^* shifts to 0.0001. The mechanism: very slow consolidation means each field update takes many calm steps to accumulate, increasing the effective persistence time of written values. As a result, even minimal turnover is sufficient to erase the sparse updates before they stabilize, pushing ν^* toward zero rather than following any proportional relationship. This is a boundary collapse, not a scaling trend, and is consistent with the operational regime boundary observed for SS above.

6 Experiment 4: Wave Input Rate (Location at Input Extremes)

6.1 Protocol

$\texttt{WAVE_RATIO} \in \{1.2, 2.4, 4.8, 9.6\}$. All other parameters at reference. $4 \times 9 \times 5 = 180$ runs.

6.2 Results

Table 4: WAVE_RATIO sweep: ν^* and $\texttt{sg4}^*$.

WR	ν^*	$\texttt{sg4}^*$	sg4 profile
1.2	0.0010	120.4	32 104 **120** 68 16 7 1 0 0
2.4	0.0010	414.9	111 328 **415** 317 34 11 2 1 0
4.8	0.0010	376.3	223 273 **376** 328 80 24 4 1 0
9.6	0.0001	226.9	**227** 70 0 28 12 1 0 0 0

Result. $\nu^* = 0.001$ for $\texttt{WR} \in \{1.2, 2.4, 4.8\}$, a 4× range. At $\texttt{WR} = 9.6$, ν^* shifts to 0.0001—*opposite* to the proportional prediction. High wave density at $\nu = 0.001$ prevents any site from achieving $\texttt{SS} = 10$ consecutive calm steps ($\texttt{sg4} = 0$); only near-zero turnover recovers partial structure. This is the over-perturbation collapse documented in Papers 7 and V93.

Notably, even WR controls location only at this extreme. Within the practical WR range $\{1.2, 2.4, 4.8\}$, ν^* is again stable: the copy-forward/erasure balance that sets the regime boundary is resilient to a 4× change in input frequency.

7 Discussion

7.1 The Invariance Finding

Across four experiments spanning propagation (κ: 0–0.15), consolidation gate (SS: 5–20), consolidation rate (FIELD_ALPHA: 0.08–0.32), and wave input rate (WR: 1.2–4.8), $\nu^* \approx 0.001$ is stable within their respective operational regimes.

Parameter	Range for ν^* stability	Effect on amplitude
κ (diffusion)	0–0.15 (full range)	Modest decrease (−9%)
SS (gate)	5–20 (4×)	Strong: 888 → 186
FIELD_ALPHA (rate)	0.08–0.32 (4×)	Proportional: 263 → 507
WR (wave rate)	1.2–4.8 (4×)	Non-monotone; peak at 2.4

The adaptive window opens at $\nu \approx 0.001$ and closes at $\nu \approx 0.007$–0.010, with these boundaries determined by the wave–grid interaction rather than the learning parameters. This is an empirical result within this substrate and input protocol; whether a similar invariance holds under qualitatively different input statistics or substrates remains an open question.

7.2 Separation of Concerns: The Core Architectural Result

The experiments distinguish two functionally independent roles:

- **Location of the adaptive regime** (ν^*, regime boundary): set by wave input dynamics. Insensitive to consolidation parameters within their operational range, and insensitive to propagation strength across the full tested range.
- **Amplitude of spatial structure** ($\texttt{sg4}^*$, peak value): strongly controlled by consolidation parameters (FIELD_ALPHA, SS) and weakly by κ. Does not alter the location where the adaptive window opens.

This is the sharpest statement the experiments support: *adaptation is local* (the learning condition is set by the per-site wave/turnover balance), while *coordination is global* (propagation amplifies already-formed local structure). The causal evidence for this separation comes from Paper 7 Experiment 6: ablating the copy-forward loop (`SEED_BETA` = 0, removing local inheritance at birth) eliminates the adaptive peak entirely, confirming that local inheritance rather than propagation is the operative mechanism.

7.3 Order-of-Magnitude Consistency Check

The observed $\nu^* \approx 0.001$ can be matched post-hoc by a simple copy-forward/erasure balance argument. At $\nu = 0.001$, each site has a mean lifetime of $1/\nu = 1000$ steps. With `WR` = 4.8 and `WAVE_DUR` = 15, approximately $4.8/15 \approx 0.32$ waves launch per step. Assuming each site is involved in half the left-zone waves and that perturbation events are approximately independent:

$$\nu^* \;\approx\; \frac{\texttt{WR}}{2 \cdot \texttt{WAVE_DUR} \cdot N_{\text{act}}/4} \;\approx\; 0.0016 \tag{4}$$

This rough estimate lands within 2× of the empirical value. It is offered as an *order-of-magnitude consistency check*, not a derived prediction: it was constructed after observing ν^*, assumes independence of perturbation events (which may not hold on the lattice), and does not account for spatial correlations or streak-length distributions. Its value is in showing which parameters plausibly set the balance (`WR`, `WAVE_DUR`, N_{act}), none of which are the consolidation or propagation parameters.

7.4 A Broader Hypothesis: Local Adaptation, Global Flow

The invariance of ν^* to κ is consistent with a broader architectural hypothesis: *the condition for effective adaptation may be local* in systems where learning occurs through mandatory turnover. If so, propagation plays a complementary but downstream role—spreading locally formed structure further, not setting the condition under which local learning succeeds.

A structurally similar pattern appears in several biological adaptive systems: hippocampal neurogenesis (local synaptic update, global pattern separation), immune adaptation (somatic hypermutation locally, clonal proliferation globally), and evolution (local mutation, global selection). In each case, the rate of local adaptation is constrained by environmental perturbation and replacement dynamics, not by the reach of global communication.

These are *possible analogies*, not mechanistic evidence. Whether the specific invariance observed here—ν^* independent of κ—generalizes to other substrates or input protocols is an open empirical question. The VCML result motivates that question; it does not answer it.

7.5 Design Implications

The separation of concerns yields a practical prescription:

- **To set the regime location**: Tune `WAVE_RATIO` and `WAVE_DUR` (the input protocol). The consistency estimate (Eq. (4)) gives the expected ν^*.
- **To increase structure amplitude**: Increase `FIELD_ALPHA` (immediate, proportional effect) or decrease `SS` (exponential P_{calm} gain, but risks over-writing at extreme values).
- **To increase spatial reach**: Increase κ (modest amplitude gain; does not shift the window location).

7.6 Amplitude Formula

The P_{calm} formula (Eq. (3)) predicts the relative consolidation rate across SS values:

SS	P_{calm} (Eq. 3)	`sg4`* (observed)
5	0.047	888
10	0.015	376
20	0.0024	186
40	6×10^{-5}	23

Observed `sg4`* drops consistently with P_{calm}, though the relationship is sub-exponential in practice (sg4 grows sub-linearly with the number of consolidation events, as sites saturate).

7.7 The Anomaly Cell Resolved

Paper 7 Experiment 3 shows `sg4` $= 68$ at $(\nu = 0.005, \kappa = 0.060)$, lower than adjacent cells at the same κ with lower ν (`sg4` $= 375$). This is not a grid artifact: $\nu = 0.005$ sits near the right boundary of the adaptive window ($\nu_{\text{crit}} \approx 0.007$–$0.010$), and the current experiments confirm that higher κ cannot rescue structure because the boundary is determined by ν, not κ. The cell at $(\nu = 0.005, \kappa = 0.060)$ is an empirical first touch of the turnover-dominated collapse region—fully consistent with the turnover law.

7.8 Limitations

The wave input protocol is fixed (zone-based, 4 zones, structured alternation). The balance equation assumes a specific wave geometry; the stability of ν^* may not hold for qualitatively different input statistics (continuous-time Poisson events, spatially diffuse perturbations). The P_{calm} formula applies an independence approximation; the true distribution accounts for spatial correlations and streak-length distributions not captured here.

8 Conclusion

An empirical turnover invariance for a VCML adaptive memory substrate: $\nu^* \approx 0.001$ is stable across the full tested range of propagation strength (κ: 0–0.15) and across 4× practical ranges of the consolidation gate (`SS`) and rate (`FIELD_ALPHA`). The regime boundary is set by the copy-forward/erasure balance at the wave–grid level; an order-of-magnitude consistency check ($\nu^* \approx$ `WR`$/(2 \cdot$ `WAVE_DUR` $\cdot N_{\text{act}}/4) \approx 0.0016$) is consistent with the empirical value but was derived post-hoc and carries caveats.

The sharpest architectural result is a separation of concerns: *adaptation is local* (the learning condition is the per-site wave/turnover balance), *coordination is global* (propagation amplifies locally formed structure but does not set the condition under which it forms). This separation, supported causally by the copy-forward ablation in Paper 7, motivates the broader hypothesis that similar invariances may appear in other adaptive systems with mandatory local turnover. Testing that hypothesis is left for future work.

A Reproducibility

Experiment 1 uses cached JSON from Paper 7. Experiments 2–4 use only `numpy`. Figure 1 is generated by `paper8_figure1_kappa_invariance.py` using the Paper 7 Exp3 cached data.

Script	Description	Runtime
`paper8_exp1_ss_sweep.py`	SS sweep (180 runs)	∼8 min
`paper8_exp2_alpha_sweep.py`	FIELD_ALPHA sweep (180 runs)	∼8 min
`paper8_exp3_wr_sweep.py`	WR sweep (180 runs)	∼8 min
`paper8_figure1_kappa_invariance.py`	Figure 1 (cached data)	<1 min

Listing 1: Run all experiments and generate figure.

```
cd papers/paper8_timescale_matching
py paper8_exp1_ss_sweep.py
py paper8_exp2_alpha_sweep.py
py paper8_exp3_wr_sweep.py
py paper8_figure1_kappa_invariance.py
```